Exp No:14

Exploring Pro-Code Extension of a Zoho Creator Application Using Zoho Catalyst / Custom Functions

Aim

To explore pro-code extensions for a Zoho Creator application using Zoho Catalyst and custom functions to implement advanced business logic and extend application capabilities.

Record Status

Completed after exploring the available custom functions and pro-code integration options.

Concept

Pro-code development involves writing code to implement functionality that goes beyond the capabilities of visual, low-code development tools.

Zoho Creator supports Deluge scripting and integrations with external services. Zoho Catalyst provides a development platform for building custom backend services and applications using supported programming languages and tools.

Combining low-code and pro-code development allows developers to build user interfaces quickly while using custom code for complex processing, external integrations, and specialized business requirements.

Procedure

1. Log in to Zoho Creator and open the Loan Application Management System.
2. Identify business requirements that need custom logic, such as advanced loan calculations or document processing.
3. Open the workflow configuration and explore the available Deluge custom-function options.

4. Create or review a custom function for processing loan application information.
5. Explore Zoho Catalyst and its supported backend development capabilities.
6. Identify how a Catalyst service could process data received from Zoho Creator through an authorized API integration.
7. Configure the required authentication and connection settings if integration is implemented.
8. Test the custom logic with sample loan application data.
9. Verify the output and confirm that the existing forms, reports, and workflows continue to work correctly.

Application Design

Case Study: Loan Application Management System

The application can be extended using the following components:

- **Zoho Creator:** Manages applicant forms, reports, dashboards, and workflows.
- **Deluge custom functions:** Perform calculations, validate inputs, and automate business rules.
- **Zoho Catalyst:** Can provide a separate backend service for advanced processing or integrations.
- **API integration:** Transfers the required information between Creator and a custom backend service.
- **Output handling:** Returns the processing result to Creator for display or further workflow actions.

```
{
   if(income <= 0)
   {
      return -1;
   }

   ratio = (liabilities / income) * 100;
   return ratio;
```

}
Example input:

- Monthly income: ₹50,000
- Existing monthly liabilities: ₹10,000
- Expected DTI: 20%

The function should be configured using the appropriate custom-function editor and supported return type in your Zoho Creator environment.

Observations

1. Custom functions support reusable business logic.
2. Deluge simplifies automation within Zoho Creator.
3. Zoho Catalyst can support additional backend processing when required.
4. API-based integration enables communication between separate services.
5. External integrations require proper authentication, error handling, and testing.

Validation and Discussion

Test the custom function using valid and invalid income values. Verify that valid inputs produce the expected DTI ratio and that zero or negative income is handled appropriately.

If a Catalyst integration is implemented, test the API request, authentication, response handling, and failure scenarios. Confirm that errors from external services do not cause incorrect updates to loan application records.

The exact integration procedure depends on the Catalyst service, programming language, authentication configuration, and Zoho Creator plan.

Result

Thus, pro-code extension techniques using Deluge custom functions and Zoho Catalyst were explored to understand how low-code applications can be extended with reusable business logic, custom backend services, and external integrations.